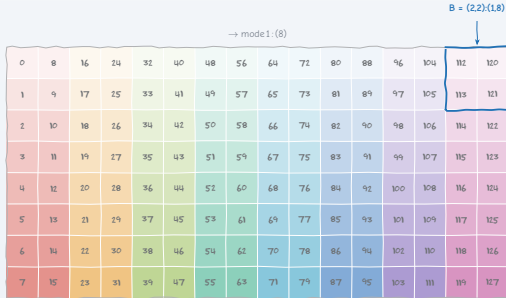


LEETCUDA

CUDA KERNEL 优化之路

从编程模型到 CUTE 与 FLASHATTENTION



L = (8,16)>(1,8)

```
auto tiled_mma = make_tiled_mma(SM80_16x8x16_F32F16F16F32_TN(), Layout<Shape<2,2,1>>());
```

郑重声明
本书仅用于开源学习与技术交流。
未经作者本人同意，不得用于任何商业用途。

LEETCUDA PROJECT (OWN BY DEFRUTHTH) · 2016 年 9 月

FA2 / FA3 ATTENTION VS CUDNN SDPA

NVIDIA RTX PRO 5000 (SM-120A) - B=1, H=32, N=16384, D=128

CONFIGURATION	TFLOPS	SPEEDUP	RELATIVE
CUDNN SDPA BASELINE	232.4	1.00X	

■ FA2 MMA

SK=2, F16ACC	157.9	0.68X	
SK=1, F32ACC	145.6	0.63X	
SK=2, F32ACC	166.5	0.72X	

■ FA2 CUTE MMA

SK=1, F32ACC	189.6	0.82X	
SK=2, F32ACC	197.0	0.85X	

■ FA2 TMA WS (1 WG)

SK=1, SV=1, F16ACC	161.2	0.69X	
SK=2, SV=1, F16ACC	189.4	0.81X	
SK=2, SV=2, F16ACC	190.5	0.82X	
SK=2, SV=1, F32ACC	204.9	0.88X	

■ FA3 TMA WS (2 WG)

SK=1, SV=1, F16ACC	210.1	0.90X	
SK=1, SV=1, F32ACC	210.8	0.91X	

■ FA2 CUTE TMA WS (1 WG)

SK=2, SV=1, F32ACC	220.0	0.95X	
SK=3, SV=1, F32ACC	223.4	0.96X	

■ FA2 CUTE TMA WS PERSISTENT-CTA (1 WG)

D=128	242.5	1.04X	
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HGEMM CUTE SWIZZLE VS CUBLAS

NVIDIA RTX PRO 5000 (SM-120A) - M=N=K=4096, F16ACC; FP8 4096/8192

CONFIGURATION	TFLOPS	SPEEDUP	RELATIVE
CUBLAS BASELINE	236.9	1.00X	

■ HGEMM CUTE SWIZZLE (F16ACC)

S=2, BLK-SW=0	229.5	0.97X	
S=2, BLK-SW=1	231.1	0.98X	
S=3, BLK-SW=0	245.6	1.04X	
S=3, BLK-SW=1	246.7	1.04X	

■ FP8 CUTE GEMM (F32ACC)

WS S2 128X256, 4096 ³	443.7	1.87X	
WS S2 128X256, 8192 ³	487.8	2.06X	

SPEEDUP IS RELATIVE TO THE VENDOR BASELINE
(CUBLAS FOR HGEMM, CUDNN SDPA FOR ATTENTION).

FA SPLIT-D VS CUDNN SDPA

NVIDIA RTX PRO 5000 (SM-120A) - B=1, H=32, N=8192, D=320

CONFIGURATION	TFLOPS	SPEEDUP	RELATIVE
CUDNN SDPA BASELINE	69.8	1.00X	

■ SPLIT-D TMA WS

SK=1, SV=1	86.6	1.24X	
SK=2, SV=2	139.9	2.01X	

■ SPLIT-D TMA NON-WS

SK=2, SV=2	187.9	2.69X	
SK=2, SV=3	188.8	2.71X	
SK=3, SV=2	204.3	2.93X	

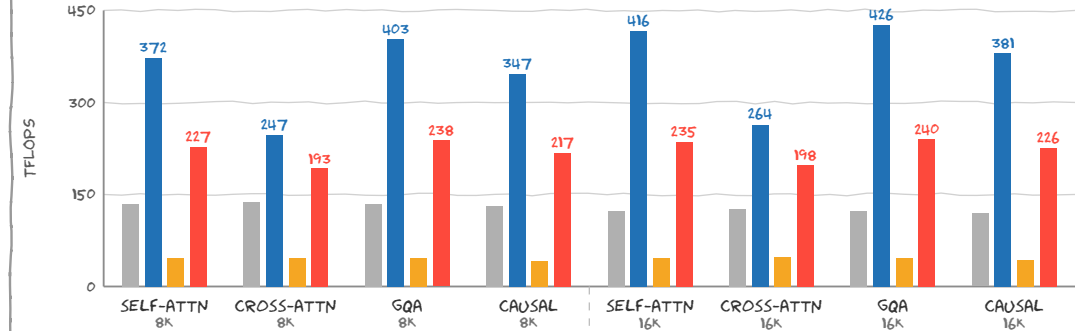
SK = K STAGES (FA), SV = V STAGES (FA), S = STAGES (HGEMM)

FFPA CUTEDSL ATTENTION (BF16, D=512) - 1.5X~15X SPEEDUP VS SDPA

FFPA CUTEDSL VS SDPA - H200 (D=512)

NVIDIA H200 (SM-90A) - B=1, H=32, D=512, BF16 - N=8192 (8K) / 16384 (16K)

■ SDPA FWD ■ FFPA FWD ■ SDPA BWD ■ FFPA BWD

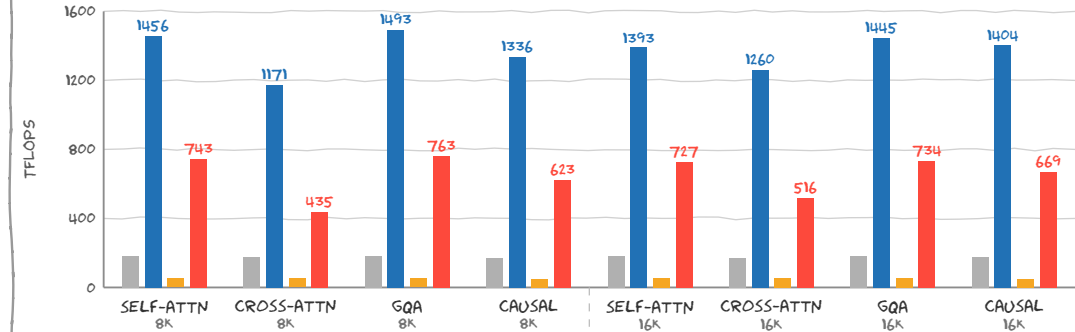


LIGHT BARS = SDPA BASELINE; FFPA BAR TOPS LABELLED IN TFLOPS. PEAK 5.2X VS SDPA.

FFPA CUTEDSL VS SDPA - B200 (D=512)

NVIDIA B200 (SM-100A) - B=1, H=32, D=512, BF16 - N=8192 (8K) / 16384 (16K)

■ SDPA FWD ■ FFPA FWD ■ SDPA BWD ■ FFPA BWD



LIGHT BARS = SDPA BASELINE; FFPA BAR TOPS LABELLED IN TFLOPS. PEAK 14.7X VS SDPA.